CS556 - Midterm Retake

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Question 1

(20 points): Your answer to any of the following questions should be 'in-sufficient information' if you feel the question does not supply enough information to determine the correct answer. Your answer can also be 'does not exist' if you judge that the question is about a property that is not applicable to the matrix in question. Given that a square matrix of dimension 45 has a linearly independent set of vectors (i.e., it is full rank), answer the following questions.

1. What is the rank of the matrix?
   The rank of the matrix is 45, since it is given that the matrix is full rank.

2. What is the numerical value of the determinant of the matrix?
   The determinant of a full rank square matrix is non-zero. However, without specific numerical values, we cannot determine the exact value of the determinant. Therefore, the answer is 'in-sufficient information'.

3. What is the rank of the column space?
   The rank of the column space is equal to the rank of the matrix, which is 45.

4. What is the dimension of the null space (i.e. the nullity)?
   The nullity of the matrix can be calculated using the formula: $\text{nullity} = \text{number of columns} - \text{rank}$ Since the matrix is square with 45 columns and has a rank of 45, the nullity is: $\text{nullity} = 45 - 45 = 0$

5. What is the dimension of the left null space (i.e. the nullity of the left null space)?
   The dimension of the left null space can be calculated using the formula: $\text{left nullity} = \text{number of rows} - \text{rank}$ Since the matrix is square with 45 rows and has a rank of 45, the left nullity is: $\text{left nullity} = 45 - 45 = 0$

6. Is the matrix invertible (why or why not)?
   The matrix is invertible because it is square and full rank, which means its determinant is non-zero.

Question 2

(20 points): Calculate the solution (if it exists) to the following system of linear equations (SLE) and demonstrate your methodology.

$x_1 - \frac{4}{7}x_2 + \frac{2}{7}x_3 = \frac{11}{7}$

$-\frac{2}{7}x_2 + \frac{1}{7}x_3 = \frac{2}{7}$

$\frac{9}{7}x_2 + \frac{6}{7}x_3 = \frac{33}{7}$

Solution:

$$\begin{bmatrix} 1 & -\frac{4}{7} & \frac{2}{7} & | & \frac{11}{7} \\ 0 & -\frac{2}{7} & \frac{1}{7} & | & \frac{2}{7} \\ 0 & \frac{9}{7} & \frac{6}{7} & | & \frac{33}{7} \end{bmatrix}$$

$$R2 = R2 + R3 \begin{bmatrix} 1 & -\frac{4}{7} & \frac{2}{7} & | & \frac{11}{7} \\ 0 & 1 & 1 & | & 5 \\ 0 & \frac{9}{7} & \frac{6}{7} & | & \frac{33}{7} \end{bmatrix}$$

$$R3 = R3 \times 7 \begin{bmatrix} 1 & -\frac{4}{7} & \frac{2}{7} & | & \frac{11}{7} \\ 0 & 1 & 1 & | & 5 \\ 0 & 9 & 6 & | & 33 \end{bmatrix}$$

$$R3 = R3 - 9R2 \begin{bmatrix} 1 & -\frac{4}{7} & \frac{2}{7} & | & \frac{11}{7} \\ 0 & 1 & 1 & | & 5 \\ 0 & 0 & -3 & | & -12 \end{bmatrix}$$

$$R3 = R3 \times -\frac{1}{3} \begin{bmatrix} 1 & -\frac{4}{7} & \frac{2}{7} & | & \frac{11}{7} \\ 0 & 1 & 1 & | & 5 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

$$R1 = \frac{4}{7}R2 + R1 \begin{bmatrix} 1 & 0 & \frac{6}{7} & | & \frac{31}{7} \\ 0 & 1 & 1 & | & 5 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

$$R2 = R2 - R3 \begin{bmatrix} 1 & 0 & \frac{6}{7} & | & \frac{31}{7} \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

$$R1 = R1 - \frac{6}{7}R3 \begin{bmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

Question 3

(15 points): Answer whether the following statements are True, False, or if there is insufficient information to make the determination. If a, b, and c are a set of linearly independent vectors, each in R^3 and none of which are the zero vector, which of the following statements are true about them?

1. a, b, and c span the space R^3.
   Since a, b, and c are linearly independent vectors in R^3 and there are three of them, they do span the space R^3. **True**

2. There exists a set of scalars λ_1, λ_2, and λ_3 where $\lambda_1 \neq \lambda_2$ which can be combined to form $\lambda_1 a + \lambda_2 b + \lambda_3 c = 0$.
   False. Since a, b, and c are linearly independent, the only solution to the equation $\lambda_1 a + \lambda_2 b + \lambda_3 c = 0$ is when all scalars are zero.

3. The determinant of the matrix with a, b, and c as columns is 0.
   False. The determinant of a matrix with linearly independent columns is non-zero.

4. Consider a specific instantiation of the vectors as $a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $b = \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$, and $c = \begin{bmatrix} -11 \\ 10 \\ 21 \end{bmatrix}$. Could this represent the set of vectors described in the original question? To determine if these vectors are linearly independent, we can form a matrix with these vectors as columns and calculate its determinant:

   $$M = \begin{bmatrix} 1 & -1 & -11 \\ 2 & 4 & 10 \\ 3 & 9 & 21 \end{bmatrix}$$

   The determinant of matrix M is calculated as follows:

   $$\det(M) = 1 \cdot (4 \cdot 21 - 9 \cdot 10) - (-1) \cdot (2 \cdot 21 - 10 \cdot 3) + (-11) \cdot (2 \cdot 9 - 4 \cdot 3)$$

   $$\det(M) = 1 \cdot (84 - 90) - (-1) \cdot (42 - 30) + (-11) \cdot (18 - 12)$$

   $$\det(M) = 1 \cdot (-6) - (-1) \cdot 12 + (-11) \cdot 6$$

   $$\det(M) = -6 + 12 - 66 = -60$$

   Since the determinant is non-zero ($-60 \neq 0$), the vectors a, b, and c are linearly independent. Therefore, this set of vectors could represent the set described in the original question. **True**

Question 4

(10 points): Calculate the inverse of 2x2 matrix $A = \begin{bmatrix} -6 & 2 \\ 9 & -3 \end{bmatrix}$.

$A^{-1} = \frac{1}{det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$

$det(A) = -6 \times -3 - 2 \times 9 = 0$

The system of equations represented by the augmented matrix has no solution. Therefore, the inverse of matrix A does not exist.

Question 5

(15 points): Employ Gram-schmidt orthogonalization to derive an orthonormal set of vectors from the set of independent vectors: $a = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

Let $A = a$

$A^T = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}$

$$AA^T = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 3$$

$$A' = \frac{A}{||A||} = \frac{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}}{\sqrt{1^2+(-1)^2+1^2}} = \frac{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}}{\sqrt{3}} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$B = b - \frac{A^T b}{A^T A} A = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$B^T = \begin{bmatrix} 1/3 & 2/3 & 1/3 \end{bmatrix}$

$$BB^T = \begin{bmatrix} 1/3 & 2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} = 2/3$$

$$B' = \frac{B}{||B||} = \frac{\begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix}}{\sqrt{2/3}} = \sqrt{\frac{3}{2}} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$$

$$A^T c = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 2, \quad B^T c = \begin{bmatrix} 1/3 & 2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \frac{5}{3}$$

$$C = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} - \frac{5/3}{2/3} \begin{bmatrix} 1/3 \\ 2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2/3 \\ -2/3 \\ 2/3 \end{bmatrix} - \begin{bmatrix} 5/6 \\ 5/3 \\ 5/6 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

$$C' = \frac{C}{||C||} = \frac{\begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}}{\sqrt{1/4+0+1/4}} = \frac{\begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}}{1/\sqrt{2}} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

The orthonormal set of vectors is:

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

Question 6

(10 points) Consider a vector $a_e = \begin{bmatrix} 3 & 1 \end{bmatrix}$ defined on the natural basis. Now, given the new basis defined by a matrix $B = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$ how is the vector a_e expressed in the new basis?

To express the vector $a_e = \begin{bmatrix} 3 & 1 \end{bmatrix}$ in the new basis defined by the matrix $B = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$, we need to find the coordinates of a_e in the new basis. This can be done by solving the equation:

$$a_e = Ba_b$$

where a_b is the vector in the new basis. To find a_b, we can use the inverse of the matrix B:

$$a_b = B^{-1}a_e$$

First, we need to find the inverse of the matrix B:

$$B = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$$

The determinant of B is:

$$\det(B) = (2)(1) - (2)(-2) = 2 + 4 = 6$$

The inverse of B is:

$$B^{-1} = \frac{1}{\det(B)} \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Now, we can find the vector a_b in the new basis:

$$a_b = B^{-1}a_e = \begin{bmatrix} \frac{1}{6} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \cdot 3 + (-\frac{1}{3}) \cdot 1 \\ \frac{1}{3} \cdot 3 + \frac{1}{3} \cdot 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - \frac{1}{3} \\ 1 + \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{4}{3} \end{bmatrix}$$

The vector $a_e = \begin{bmatrix} 3 & 1 \end{bmatrix}$ expressed in the new basis defined by the matrix $B = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$ is:

$$a_b = \begin{bmatrix} \frac{1}{6} \\ \frac{4}{3} \end{bmatrix}$$

Question 7

(10 points) Find all the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$.

To find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$, we follow these steps: The eigenvalues λ of matrix A are found by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$

where I is the identity matrix.

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix}$$

The determinant of $A - \lambda I$ is:

$$\det(A - \lambda I) = (1-\lambda)(3-\lambda) - (2)(4) = (3 - \lambda - 3\lambda + \lambda^2) - 8 = \lambda^2 - 4\lambda - 5$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda = \frac{4 \pm \sqrt{16+20}}{2} = \frac{4 \pm 6}{2}$$

The eigenvalues are:

$$\lambda_1 = 5 \quad \text{and} \quad \lambda_2 = -1$$

For each eigenvalue, we find the corresponding eigenvector by solving $(A - \lambda I)v = 0$.

For $\lambda_1 = 5$:

$$A - 5I = \begin{bmatrix} 1-5 & 2 \\ 4 & 3-5 \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix}$$

The system of equations is:

$$-4x + 2y = 0$$

$$2x = y$$

The eigenvectors:

$$v_1 = \begin{bmatrix} x \\ 2x \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $x = 1$, we get:

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $\lambda_2 = -1$:

$$A - (-1)I = \begin{bmatrix} 1-(-1) & 2 \\ 4 & 3-(-1) \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix}$$

$$2x + 2y = 0$$

$$x = -y$$

Thus, the eigenvector can be written as:

$$v_2 = \begin{bmatrix} x \\ -x \end{bmatrix} = x \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For $x = 1$, we get:

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The eigenvectors are:

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The eigenvalues of the matrix A are $\boxed{5}$ and $\boxed{-1}$, with corresponding eigenvectors $\boxed{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}$ and $\boxed{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}$, respectively.

Question 8

(5 Points) Find the best fit line using least squares approximation for the points (1, 2), (2, 1), (3, 1) in R^2

Solution

$A^T A\hat{x} = A^T b$

$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$

$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix}$

$A^T b = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$

$\hat{x} = (A^T A)^{-1} A^T b$

$\hat{x} = (\begin{bmatrix} 14 & 6 \\ 6 & 3 \end{bmatrix})^{-1} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.5 \\ \frac{7}{3} \end{bmatrix}$

The best fit line is $y = 0.5x + \frac{7}{3}$